



Sydney Girls High School

2021 Alternate Task 4

Mathematics Extension 1

General

Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using a black pen
- Calculators approved by NESA may be used
- A NESA reference sheet has been provided for use
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks :

40

Section 1 – 10 marks (pages 2 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 30 marks (pages 7 – 12)

- Attempt Questions **11 – 13**
- Allow about 45 minutes for this section

THIS IS NOT A TRIAL PAPER

It does not reflect the format or the content of the 2021 HSC Examination paper
in this subject.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

For each question, select the correct response A, B, C or D.

List the correct response only on YOUR writing paper for questions 1 – 10.

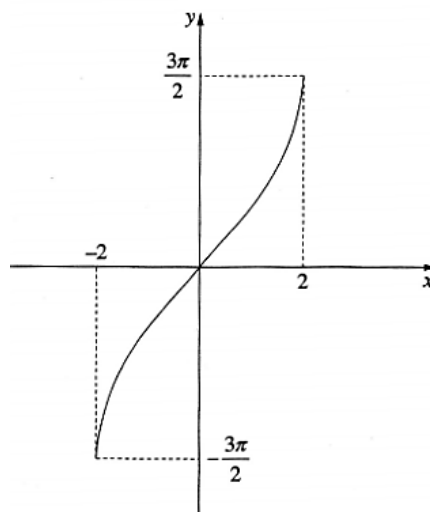
1. Which function best describes the following graph?

(A) $y = 3 \sin^{-1} 2x$

(B) $y = \frac{3}{2} \sin^{-1} 2x$

(C) $y = 3 \sin^{-1} \frac{x}{2}$

(D) $y = \frac{3}{2} \sin^{-1} \frac{x}{2}$



2. In the cartesian plane, a vector perpendicular to the line $3x + 2y + 1 = 0$ is

(A) $3\vec{i} + 2\vec{j}$

(B) $-\frac{1}{2}\vec{i} + \frac{1}{3}\vec{j}$

(C) $2\vec{i} - 3\vec{j}$

(D) $\frac{1}{2}\vec{i} - \frac{1}{3}\vec{j}$

3. Which group of three numbers could be the roots of the polynomial equation

$$x^3 + px^2 - 26x + 24 = 0 ?$$

(A) 2, 3, 4

(B) 1, -6, 4

(C) -1, -2, 12

(D) -1, -3, -8

4.
$$\frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} =$$

(A) $\cos \theta$

(B) $\sec \theta$

(C) $\tan \theta$

(D) $\cot \theta$

5. Which expression is equal to $\int \cos^2 \frac{2x}{5} dx$?

(A) $\frac{x}{2} - \frac{2}{5} \sin \frac{4x}{5} + C$

(B) $\frac{x}{2} + \frac{2}{5} \sin \frac{4x}{5} + C$

(C) $\frac{x}{2} - \frac{5}{8} \sin \frac{4x}{5} + C$

(D) $\frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + C$

6. In the expression $(2x + k)^6$ the coefficients of x and x^2 are equal.

What is the value of k ?

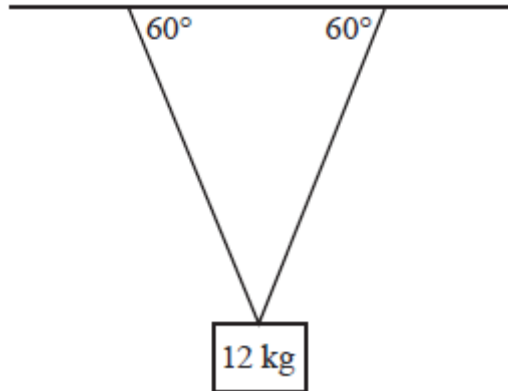
- (A) 5
- (B) 6
- (C) 11
- (D) 12

7. A group consisting of two adults, two boys and two girls is to be seated at a round table. The adults are to be seated together. The boys and girls are to sit in alternating positions? How many different seating arrangements are possible?

- (A) 8
- (B) 16
- (C) 24
- (D) 30

8. A 12kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of 60° with the ceiling, as shown. If the force of gravity is $g \text{ m/s}^2$ then the magnitude, in newtons, of the tension in each string is equal to:

- (A) $12g$
(B) $24g$
(C) $4\sqrt{3}g$
(D) $8\sqrt{3}g$

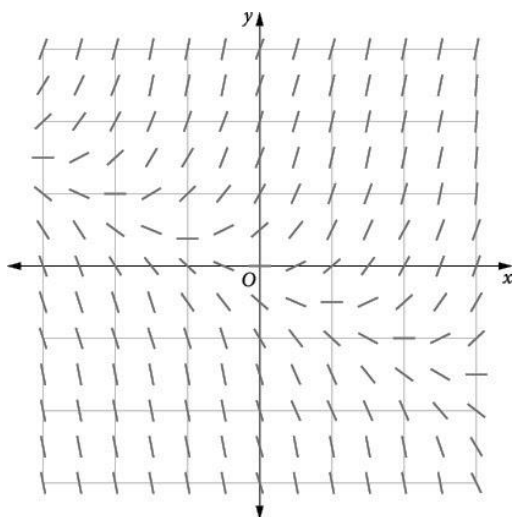


9. The volume V of a spherical balloon of radius r mm is increasing at a constant rate of 800 mm^3 per second. What is the rate of change of the radius with respect to time?

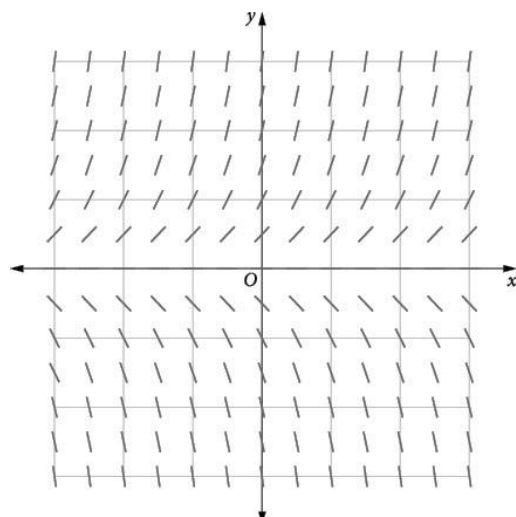
- (A) $\frac{\pi r}{100} \text{ mm/second}$
(B) $\frac{100}{\pi r} \text{ mm/second}$
(C) $\frac{\pi r^2}{200} \text{ mm/second}$
(D) $\frac{200}{\pi r^2} \text{ mm/second}$

10. Which of the following direction fields could have the differential equation $\frac{dy}{dx} = x - ky$ as a solution if $k > 0$?

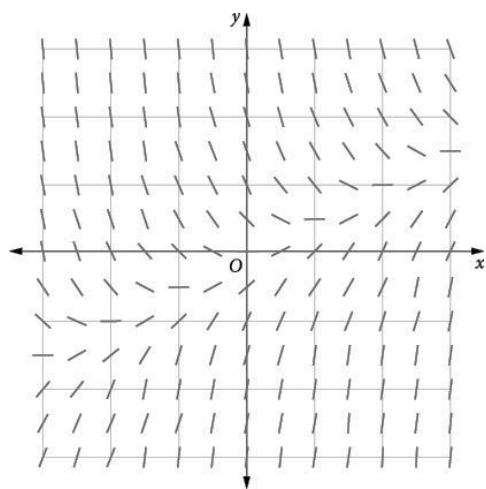
(A)



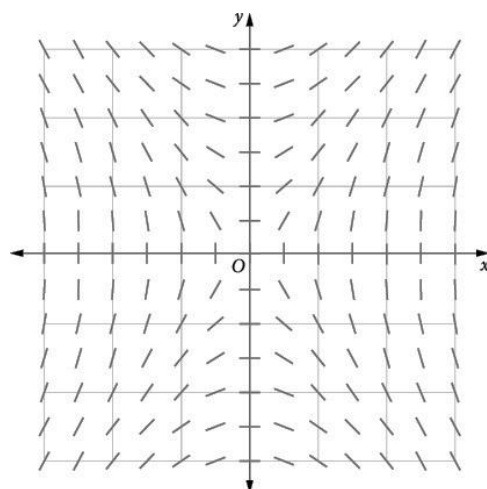
(B)



(C)



(D)



Section II

30 marks

Attempt Questions 11 – 13

Allow about 45 minutes for this section

Begin a new page for each question.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 Begin a new Page

(10 marks)

(a)

(i) Show that $(x - 3)$ is a factor of the polynomial

1

$$P(x) = x^3 - 4x^2 + x + 6.$$

(ii) Hence, express $P(x)$ in factored form.

2

(b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{4x} \right)$

1

(c) Solve: $\frac{x^2 - 9}{3x} > 0$

2

(d) Evaluate:

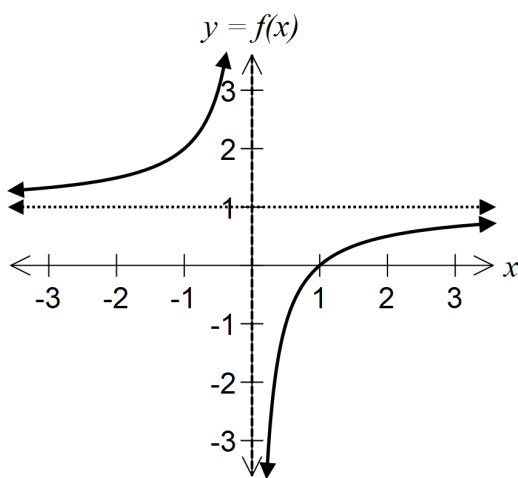
2

$$\int \frac{2x}{\sqrt{25 - x^2}} dx$$

Question 11 continues on the next page.

Question 11 (continued)

(e) Consider the function $f(x) = \frac{x-1}{x}$, which is graphed below:



Using the graph of $y = f(x)$, sketch the function $y = \{f(x)\}^2$.

2

End of Question 11

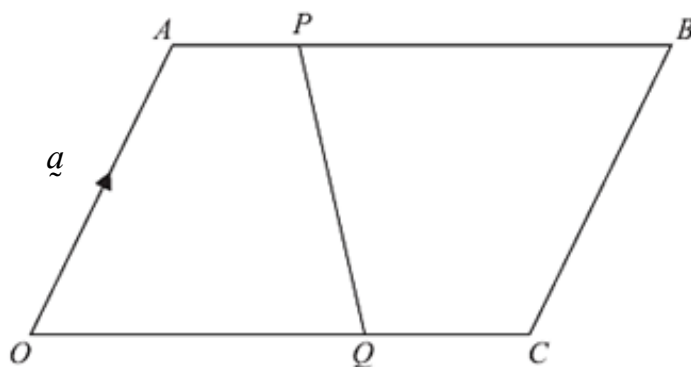
Question 12 Begin a new page.**(10 marks)**

- (a) In the diagram below, $OABC$ is a parallelogram. $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OC} = \underline{c}$ **3**

P is the point on AB such that $AP = \frac{1}{4}AB$.

Q is the point on OC such that $OQ = \frac{2}{3}OC$.

Find $\overrightarrow{PQ}$ in terms of $\underline{a}$ and $\underline{c}$, giving your answer in simplest form.



- (b) Find the volume, in terms of π , of the solid formed when the area **3**
bounded by the curve $y = \frac{1}{2}\log_e x$, the lines $y = -1$, $y = 2$ and the y -axis
is rotated about the y -axis.

Question 12 continues on the next page.

Question 12 (continued)

(c) Consider the expansion of $(1+x)^{n-1}$ for integers $n > 2$.

(i) Show that:

$$n\binom{n-1}{1} + n\binom{n-1}{2} + \dots + n\binom{n-1}{n-2} = n(2^{n-1} - 2) \quad \mathbf{3}$$

(ii) Find the smallest positive integer n such that:

$$n\binom{n-1}{1} + n\binom{n-1}{2} + \dots + n\binom{n-1}{n-2} > 5000 \quad \mathbf{1}$$

End of Question 12

Question 13 Begin a new page.**(10 marks)**

- (a) A particle is moving in a straight line and is oscillating backwards and forwards. At time t seconds, it has displacement x metres from a fixed point O on the line, where $x = A \cos\left(\frac{\pi}{4}t + \alpha\right)$, $A > 0$, and $0 < \alpha < \frac{\pi}{2}$. After 1 second the particle is 2 metres to the right of O , and after 3 seconds it is 4 metres to the left of O .

Given that $A \cos \alpha - A \sin \alpha = 2\sqrt{2}$ and $A \cos \alpha + A \sin \alpha = 4\sqrt{2}$:

- (i) Solve the equations simultaneously, to find the values of A and α in exact form. **2**
- (ii) Show that the particle first passes through O after $\frac{4}{\pi} \tan^{-1} 3$ seconds. **2**

Question 13 continues on the next page.

Question 13 (continued)

- (b) The position coordinates of any point on the path of a projectile at time $t \geq 0$ in seconds, with initial velocity $v \text{ ms}^{-1}$ at an angle of projection θ , and acceleration downwards due to gravity, g , are:

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - \frac{1}{2}gt^2$$

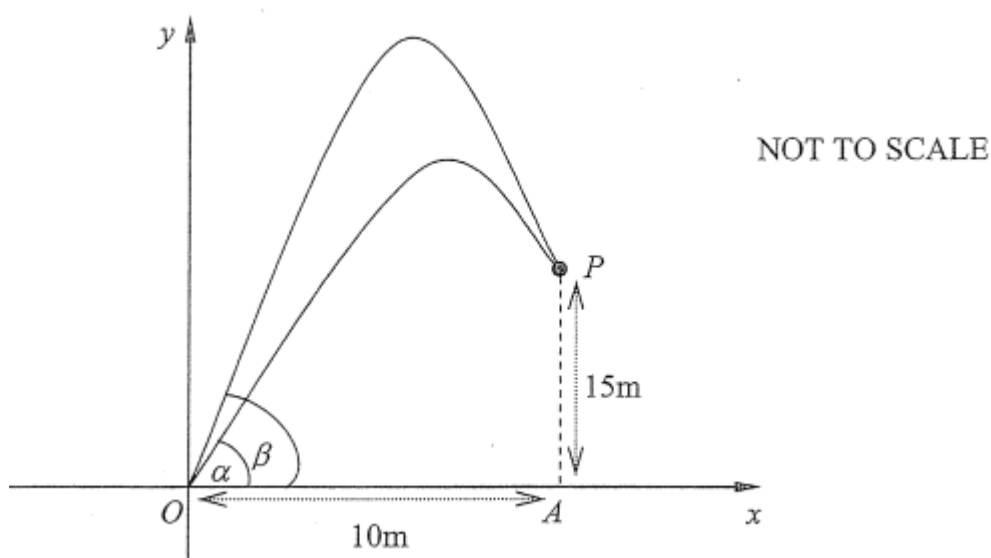
- (i) Show that the equation of the path of a projectile is given by:

2

$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

Harry throws a tennis ball from a fixed point O on level ground, with a velocity $v = 7\sqrt{10} \text{ ms}^{-1}$ at an angle β with the horizontal. Shortly afterwards he throws another tennis ball from the same point at the same speed but at a different angle to the horizontal, α , where $\alpha < \beta$ as shown.

The two tennis balls collide at a point P , vertically above the point A on the ground, where $OA = 10 \text{ m}$ and $AP = 15 \text{ m}$. The acceleration downwards due to gravity is $g = 9.8 \text{ ms}^{-2}$.



- (ii) Show that $\tan \alpha = 2$ and $\tan \beta = 8$.

2

- (iii) Find, in surd form, the time elapsed between when the tennis balls were thrown.

2

End of paper



SOLUTIONS

Sydney Girls High School

2021 Alternate Task 4

Mathematics Extension 1

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in this subject.

2021 Extension 1 Task 4

Multiple choice

1. C

6. A

2. A

7. B

3. B

8. C

4. A

9. D

5. D

10. C

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

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$$D: -1 \leq \left(\frac{x}{2}\right) \leq 1$$

$$-2 \leq x \leq 2$$

$$R: -\frac{\pi}{2} \leq \left(\frac{y}{3}\right) \leq \frac{\pi}{2}$$

$$-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$$

$$\therefore \frac{y}{3} = \sin^{-1} \frac{x}{2}$$

$$y = 3 \sin^{-1} \frac{x}{2}$$

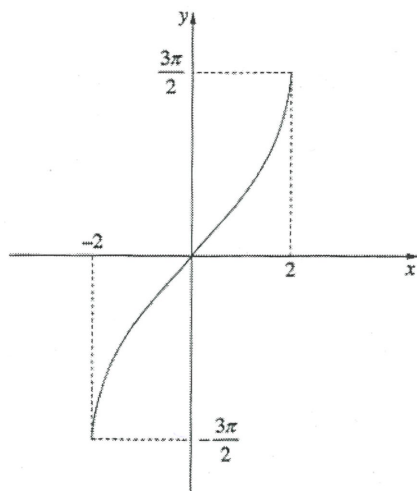
1. Which function best describes the following graph?

(A) $y = 3 \sin^{-1} 2x$

(B) $y = \frac{3}{2} \sin^{-1} 2x$

(C) $y = 3 \sin^{-1} \frac{x}{2}$

(D) $y = \frac{3}{2} \sin^{-1} \frac{x}{2}$



2. In the cartesian plane, a vector perpendicular to the line $3x + 2y + 1 = 0$ is

(A) $3\vec{i} + 2\vec{j}$

(B) $-\frac{1}{2}\vec{i} + \frac{1}{3}\vec{j}$

(C) $2\vec{i} - 3\vec{j}$

(D) $\frac{1}{2}\vec{i} - \frac{1}{3}\vec{j}$

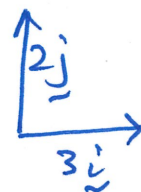
$$3x + 2y + 1 = 0:$$

$$2y = -3x - 1$$

$$y = -\frac{3}{2}x - \frac{1}{2}$$

$$m_L = -\frac{3}{2}$$

$$\therefore m_V = \frac{2}{3}$$



$$\therefore 3\vec{i} + 2\vec{j}$$

3. Which group of three numbers could be the roots of the polynomial equation

$$x^3 + px^2 - 26x + 24 = 0$$

$$x^3 + \overset{-b}{p}x^2 + \overset{+c}{-26}x + \overset{-d}{24} = 0$$

(A) 2, 3, 4

(B) 1, -6, 4

(C) -1, -2, 12

(D) -1, -3, -8

$$\alpha\beta\gamma = -24$$

$$2 \times 3 \times 4 \neq -24 \therefore \text{not (A)}$$

$$-1 \times -2 \times 12 \neq -24 \therefore \text{not (C)}$$

$$\alpha\beta + \beta\gamma + \alpha\gamma = -26$$

$$1 \times -6 + 1 \times 4 + -6 \times 4 = -6 + 4 - 24 = -26 \therefore \text{(B)}$$

4.

$$\frac{\cot \frac{\theta}{2} - \tan \frac{\theta}{2}}{\cot \frac{\theta}{2} + \tan \frac{\theta}{2}} =$$

(A) $\cos \theta$

(B) $\sec \theta$

(C) $\tan \theta$

(D) $\cot \theta$

$$\text{Let } t = \tan \frac{\theta}{2}$$

$$\Rightarrow \frac{\frac{1}{t} - t}{\frac{1}{t} + t}$$

$$= \frac{1 - t^2}{t} \div \frac{1 + t^2}{t}$$

$$= \frac{1 - t^2}{t} \times \frac{t}{1 + t^2}$$

$$= \frac{1 - t^2}{1 + t^2} = \cos \theta \quad \text{(A)}$$

5. Which expression is equal to $\int \cos^2 \frac{2x}{5} dx$?

(A) $\frac{x}{2} - \frac{2}{5} \sin \frac{4x}{5} + C$

(B) $\frac{x}{2} + \frac{2}{5} \sin \frac{4x}{5} + C$

(C) $\frac{x}{2} - \frac{5}{8} \sin \frac{4x}{5} + C$

(D) $\frac{x}{2} + \frac{5}{8} \sin \frac{4x}{5} + C$

$$\cos^2 x = \frac{1}{2} + \frac{1}{2} \cos 2x$$

$$\cos^2 \left(\frac{2x}{5} \right) = \frac{1}{2} + \frac{1}{2} \cos \left(2 \times \frac{2x}{5} \right)$$

$$\cos^2 \left(\frac{2x}{5} \right) = \frac{1}{2} + \frac{1}{2} \cos \left(\frac{4x}{5} \right)$$

$$\therefore \int \left(\frac{1}{2} + \frac{1}{2} \cos \left(\frac{4x}{5} \right) \right) dx$$

$$= \frac{1}{2}x + \frac{1}{2} \times \frac{5}{4} \sin \frac{4x}{5}$$

$$= \frac{1}{2}x + \frac{5}{8} \sin \frac{4x}{5} + C$$

6. In the expression $(2x + k)^6$ the coefficients of x and x^2 are equal.

What is the value of k ?

(A) 5

(B) 6

(C) 11

(D) 12

$$(2x + k)^6 = {}^6C_0 (2x)^6 + {}^6C_1 (2x)^5 k + {}^6C_2 (2x)^4 k^2 \\ + {}^6C_3 (2x)^3 k^3 + {}^6C_4 (2x)^2 k^4 + {}^6C_5 (2x) k^5 + {}^6C_6 (2x)^0 k^6$$

$$\text{coeff of } x = \text{coeff of } x^2$$

$${}^6C_5 \times 2 \times k^5 = {}^6C_4 \times 2^2 \times k^4$$

$${}^6C_5 \times k = {}^6C_4 \times 2$$

$$k = \frac{{}^6C_4 \times 2}{{}^6C_5} = \frac{30}{6} \therefore k = 5$$

7. A group consisting of two adults, two boys and two girls is to be seated at a round table. The adults are to be seated together. The boys and girls are to sit in alternating positions? How many different seating arrangements are possible?

2 Adults
2 girls
2 boys
6

(A) 8

(B) 16

(C) 24

(D) 30

• Seat the first adult anywhere .

• 2nd adult has 2 ways .

• seat a first boy has 4 ways .

• 2nd boy has only 1 way .

• two girls sit 2 ways

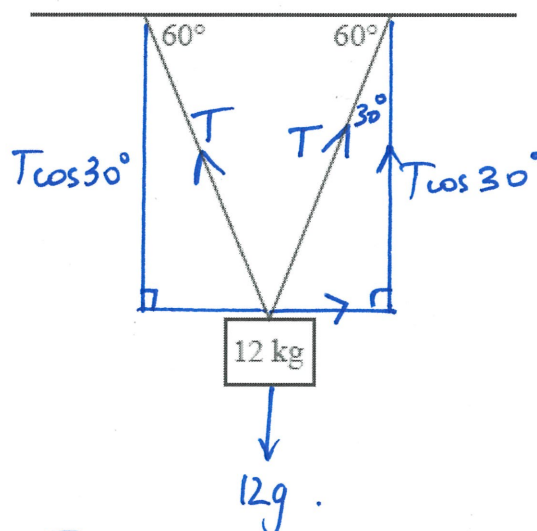
$\therefore$ seating arrangements

$$= 2 \times 4 \times 2$$

$$= 16$$

8. A 12kg mass is suspended in equilibrium from a horizontal ceiling by two identical light strings. Each string makes an angle of 60° with the ceiling, as shown. If the force of gravity is $g \text{ m/s}^2$ then the magnitude, in newtons, of the tension in each string is equal to:

- (A) $12g$
 (B) $24g$
 (C) $4\sqrt{3}g$
 (D) $8\sqrt{3}g$



$$2T \cos 30^\circ = 12g$$

$$T = \frac{12g}{2 \cos 30^\circ}$$

$$T = \frac{12g}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 4\sqrt{3}g$$

9. The volume V of a spherical balloon of radius r mm is increasing at a constant rate of 800 mm^3 per second. What is the rate of change of the radius with respect to time?

(A) $\frac{\pi r}{100} \text{ mm/second}$

(B) $\frac{100}{\pi r} \text{ mm/second}$

(C) $\frac{\pi r^2}{200} \text{ mm/second}$

(D) $\frac{200}{\pi r^2} \text{ mm/second}$

$\frac{dV}{dt} = 800 \text{ mm}^3/\text{second}$. Find $\frac{dr}{dt}$.

$$V = \frac{4}{3}\pi r^3$$

$$\frac{dV}{dr} = 4\pi r^2$$

$$\therefore \frac{dr}{dt} = \frac{dV}{dt} \times \frac{dr}{dV}$$

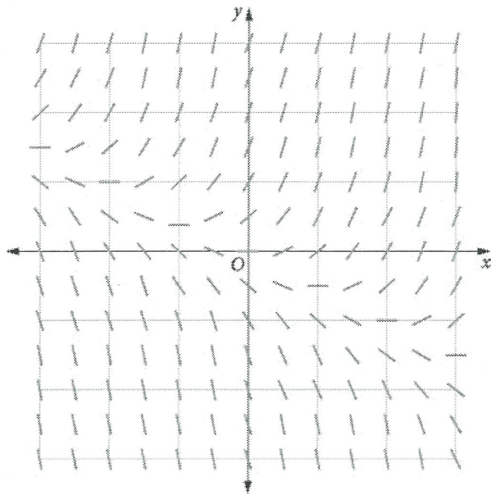
$$= 800 \times \frac{1}{4\pi r^2}$$

$$= \frac{200}{\pi r^2} \text{ mm/s}$$

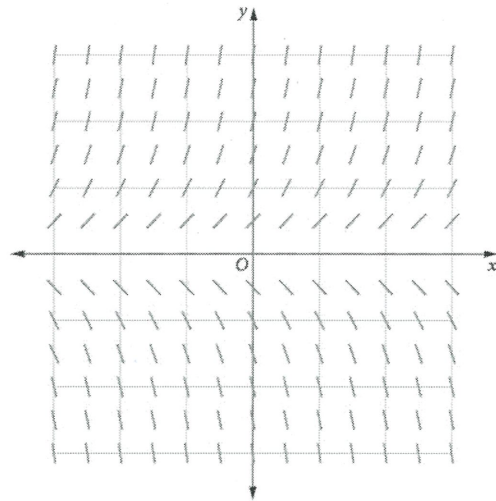
10. Which of the following direction fields could have the differential equation

$$\frac{dy}{dx} = x - ky \text{ as a solution if } k > 0 ?$$

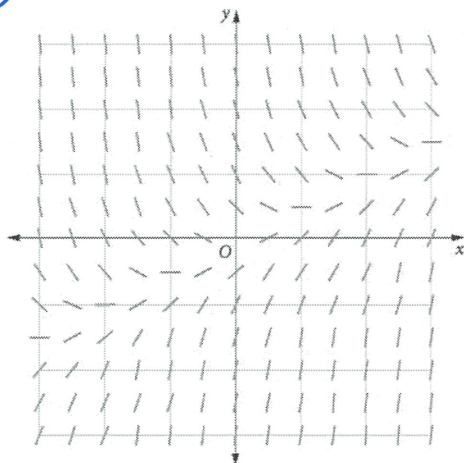
(A)



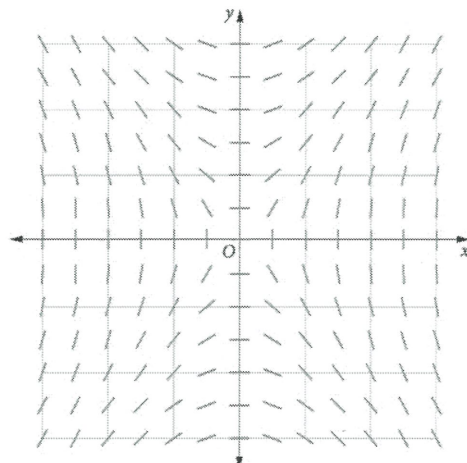
(B)



(C)



(D)



when $\frac{dy}{dx} = 0$: $ky = x \Rightarrow y = \frac{x}{k}$ (line through origin, having horizontal line elements.
 $\therefore$ not (B) and not (D).

But $k > 0$, $\therefore y = \frac{x}{k} \swarrow \therefore$ (C).

(a)

(i) Show that $(x-3)$ is a factor of the polynomial

1

$$P(x) = x^3 - 4x^2 + x + 6.$$

(ii) Hence, express $P(x)$ in factored form.

2

$$i) \quad P(3) = 3^3 - 4(3)^2 + 3 + 6$$

$$\therefore P(3) = 0. \quad \checkmark$$

①

$$ii) \quad \begin{array}{r} x^2 - x - 2 \\ x-3 \overline{) x^3 - 4x^2 + x + 6} \\ \underline{x^3 - 3x^2} \\ -x^2 + x \\ \underline{-x^2 + 3x} \\ -2x + 6 \\ \underline{-2x + 6} \\ 0 \end{array}$$

① for working if answer is incorrect.

$$\therefore P(x) = (x-3)(x^2 - x - 2)$$

$$\underline{P(x) = (x-3)(x-2)(x+1)} \quad \checkmark \checkmark \quad \textcircled{2}$$

(b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{4x} \right)$

1

$$\lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{4x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{\frac{x}{3}} \right) \times \frac{(\frac{x}{3})}{4x}$$

$$= 1 \times \frac{\cancel{x}}{3} \times \frac{1}{4\cancel{x}}$$

$$= \frac{1}{12} \quad \checkmark$$

①

(c) Solve: $\frac{x^2-9}{3x} > 0$

2

$$\frac{x^2-9}{3x} > 0$$

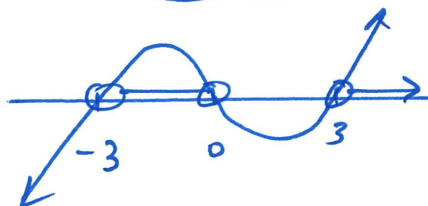
$$3x^2 \times \frac{x^2-9}{3x} > 0 \times 3x^2$$

$$x(x-3)(x+3) > 0$$

By observation:

$$\underline{\underline{-3 < x < 0 \text{ and } x > 3}} \quad \checkmark \checkmark$$

① for working
if answer is
incorrect



②.

(d) Evaluate:

2

$$\int \frac{2x}{\sqrt{25-x^2}} dx$$

$$\int \frac{2x}{\sqrt{25-x^2}} dx$$

$$= \int 2x \cdot (25-x^2)^{-\frac{1}{2}} dx \quad \checkmark$$

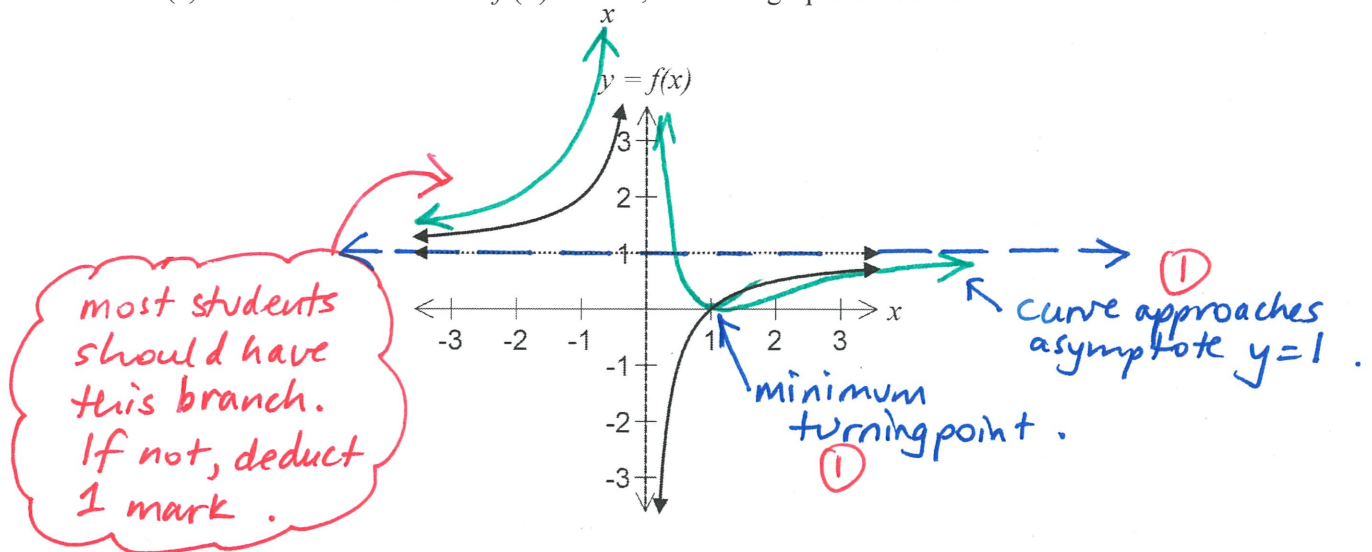
$$= -\frac{1}{\frac{1}{2}} (25-x^2)^{\frac{1}{2}} + C \quad (\text{reverse chain rule})$$

$$= \underline{\underline{-2\sqrt{25-x^2} + C}} \quad \checkmark$$

②.

Question 11 (continued)

(e) Consider the function $f(x) = \frac{x-1}{x}$, which is graphed below:



Using the graph of $y = f(x)$, sketch the function $y = \{f(x)\}^2$.

2

End of Question 11

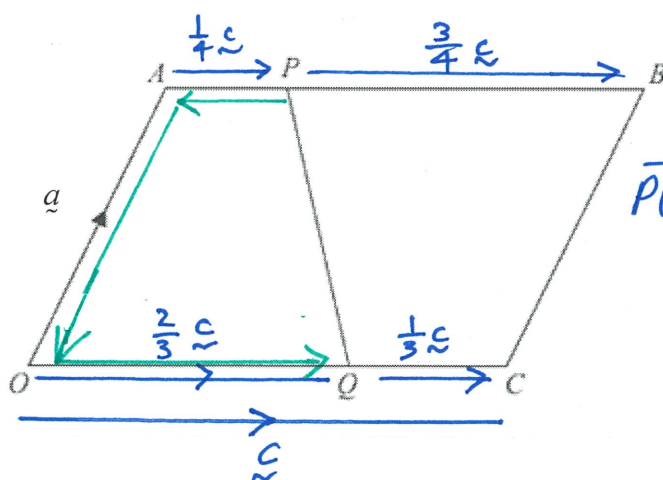
(a) In the diagram below, $OABC$ is a parallelogram. $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OC} = \underline{c}$

3

P is the point on AB such that $AP = \frac{1}{4}AB$.

Q is the point on OC such that $OQ = \frac{2}{3}OC$.

Find $\overrightarrow{PQ}$ in terms of $\underline{a}$ and $\underline{c}$, giving your answer in simplest form.



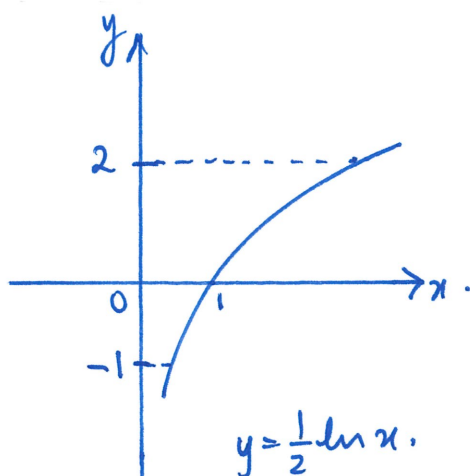
$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{PA} + \overrightarrow{AO} + \overrightarrow{OQ} \quad \checkmark \\ &= -\frac{1}{4}\underline{c} - \underline{a} + \frac{2}{3}\underline{c} \quad \checkmark \\ &= \left(\frac{2}{3} - \frac{1}{4}\right)\underline{c} - \underline{a} \\ &= \frac{5}{12}\underline{c} - \underline{a} \quad \checkmark \quad (3)\end{aligned}$$

(b) Find the volume, in terms of π , of the solid formed when the area

3

bounded by the curve $y = \frac{1}{2}\log_e x$, the lines $y = -1$, $y = 2$ and the y -axis

is rotated about the y -axis.



$$\begin{aligned}y &= \frac{1}{2}\ln x \\ 2y &= \ln x \\ e^{2y} &= x \quad \checkmark\end{aligned}$$

$$\begin{aligned}V &= \pi \int_a^b x^2 dy \\ &= \pi \int_{-1}^2 (e^{2y})^2 dy \quad \checkmark \\ &= \pi \int_{-1}^2 e^{4y} dy \\ &= \left[\frac{e^{4y}}{4} \right]_{-1}^2 \times \pi \\ V &= \left(\frac{e^8}{4} - \frac{e^{-4}}{4} \right) \pi \text{ units}^3 \quad \checkmark\end{aligned}$$

(3)

Question 12 (continued)

(c) Consider the expansion of $(1+x)^{n-1}$ for integers $n > 2$.

(i) Show that:

$$n \binom{n-1}{1} + n \binom{n-1}{2} + \dots + n \binom{n-1}{n-2} = n(2^{n-1} - 2)$$

3

(ii) Find the smallest positive integer n such that:

$$n \binom{n-1}{1} + n \binom{n-1}{2} + \dots + n \binom{n-1}{n-2} > 5000$$

1

$$i) (1+x)^{n-1} = 1 + \binom{n-1}{1}x + \binom{n-1}{2}x^2 + \dots + \binom{n-1}{n-2}x^{n-2} + x^{n-1}$$

Let $x=1$:

$$(1+1)^{n-1} = 1 + \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} + 1 \quad \checkmark$$

$$2^{n-1} = 2 + \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2}$$

$$2^{n-1} - 2 = \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} \quad \checkmark$$

$$n(2^{n-1} - 2) = n \left[\binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-2} \right] \quad \checkmark$$

$$\therefore n \binom{n-1}{1} + n \binom{n-1}{2} + n \binom{n-1}{3} + \dots + n \binom{n-1}{n-2} = n(2^{n-1} - 2).$$

③

$$ii) \text{ Solve } n(2^{n-1} - 2) > 5000$$

Use trial and error:

$$n=8: \quad 8(2^7 - 2) = 1008$$

$$n=9: \quad 9(2^8 - 2) = 2286$$

$$n=10: \quad 10(2^9 - 2) = 5100 \quad \therefore \underline{n=10} \quad \checkmark$$

①

Question 13 Begin a new page.**(10 marks)**

- (a) A particle is moving in a straight line and is oscillating backwards and forwards. At time t seconds, it has displacement x metres from a fixed point O on the line, where $x = A \cos\left(\frac{\pi}{4}t + \alpha\right)$, $A > 0$, and $0 < \alpha < \frac{\pi}{2}$.
After 1 second the particle is 2 metres to the right of O , and after 3 seconds it is 4 metres to the left of O .

Given that $A \cos \alpha - A \sin \alpha = 2\sqrt{2}$ and $A \cos \alpha + A \sin \alpha = 4\sqrt{2}$:

- (i) Solve the equations simultaneously, to find the values of A and α in exact form. 2

- (ii) Show that the particle first passes through O after $\frac{4}{\pi} \tan^{-1} 3$ seconds. 2

i) $A \cos \alpha - A \sin \alpha = 2\sqrt{2} \dots \textcircled{1}$

$A \cos \alpha + A \sin \alpha = 4\sqrt{2} \dots \textcircled{2}$

$\textcircled{1} + \textcircled{2} : 2A \cos \alpha = 6\sqrt{2} \Rightarrow A \cos \alpha = 3\sqrt{2}$

$\textcircled{2} - \textcircled{1} : 2A \sin \alpha = 2\sqrt{2} \Rightarrow A \sin \alpha = \sqrt{2}$

$\therefore A^2 (\cos^2 \alpha + \sin^2 \alpha) = (3\sqrt{2})^2 + (\sqrt{2})^2$

$A^2 = 20$

$A = 2\sqrt{5}, A > 0$. ✓

$\frac{A \sin \alpha}{A \cos \alpha} = \frac{\sqrt{2}}{3\sqrt{2}}$

$\therefore \tan \alpha = \frac{1}{3}$

$\alpha = \tan^{-1} \frac{1}{3}$ ($0 < \alpha < \frac{\pi}{2}$) ✓ 2.

ii) when $x=0$: $\cos\left(\frac{\pi}{4}t + \alpha\right) = 0$

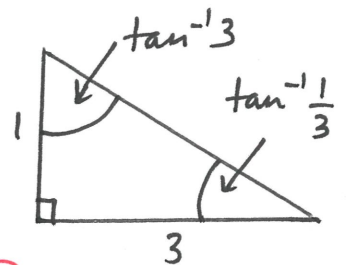
$t > 0$: $\frac{\pi}{4}t + \alpha = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

First such $t > 0$: $\frac{\pi}{4}t = \frac{\pi}{2} - \alpha$ ①

$\frac{\pi}{4}t = \frac{\pi}{2} - \tan^{-1}\frac{1}{3}$

$\frac{\pi}{4}t = \tan^{-1}3$ ①

$\therefore t = \frac{4}{\pi} \tan^{-1}3$



Question 13 (continued)

- (b) The position coordinates of any point on the path of a projectile at time $t \geq 0$ in seconds, with initial velocity $v \text{ ms}^{-1}$ at an angle of projection θ , and acceleration downwards due to gravity, g , are:

$$x = vt \cos \theta \quad \text{and} \quad y = vt \sin \theta - \frac{1}{2}gt^2$$

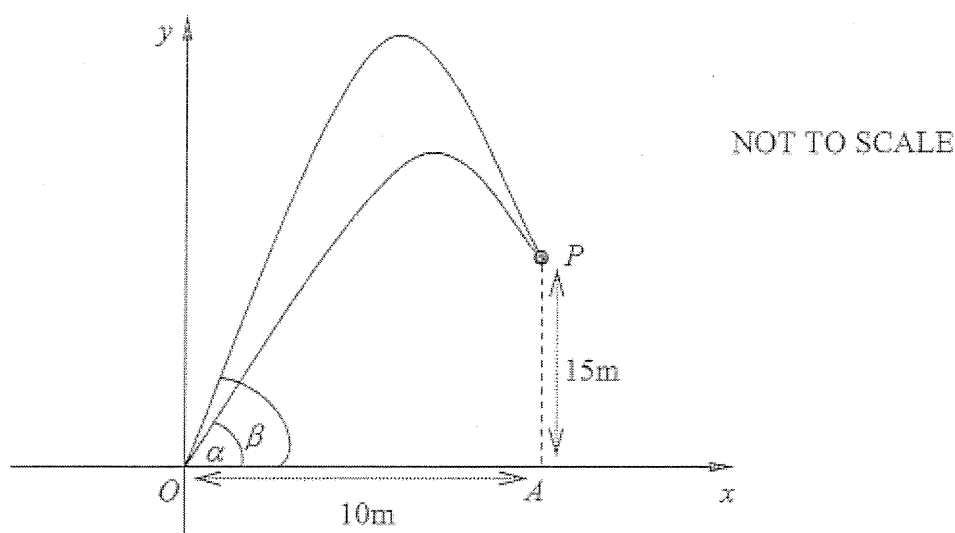
- (i) Show that the equation of the path of a projectile is given by:

$$y = x \tan \theta - \frac{gx^2}{2v^2} \sec^2 \theta$$

2

Harry throws a tennis ball from a fixed point O on level ground, with a velocity $v = 7\sqrt{10} \text{ ms}^{-1}$ at an angle β with the horizontal. Shortly afterwards he throws another tennis ball from the same point at the same speed but at a different angle to the horizontal, α , where $\alpha < \beta$ as shown.

The two tennis balls collide at a point P , vertically above the point A on the ground, where $OA = 10 \text{ m}$ and $AP = 15 \text{ m}$. The acceleration downwards due to gravity is $g = 9.8 \text{ ms}^{-2}$.



- (ii) Show that $\tan \alpha = 2$ and $\tan \beta = 8$.
- (iii) Find, in surd form, the time elapsed between when the tennis balls were thrown.

2

2

End of paper

b) i) $x = vt \cos \theta$ and $y = vt \sin \theta - \frac{1}{2} g t^2$

$\Rightarrow t = \frac{x}{v \cos \theta}$ sub into y :

$$y = (v \sin \theta) \cdot \frac{x}{v \cos \theta} - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2 \quad \checkmark$$

$$y = x \tan \theta - \frac{g x^2}{2 v^2 \cos^2 \theta}$$

$$y = x \tan \theta - \frac{g x^2}{2 v^2} \sec^2 \theta \quad \checkmark \quad (2)$$

ii) At P: $x=10, y=15$ and $g=9.8, v=7\sqrt{10}$.

Using i): $15 = 10 \tan \theta - \frac{9.8(10)^2}{2(7\sqrt{10})^2} \sec^2 \theta$

$$15 = 10 \tan \theta - \frac{9.8 \times 100}{2 \times 49 \times 10} (1 + \tan^2 \theta) \quad \checkmark$$

$$15 = 10 \tan \theta - 1 - \tan^2 \theta$$

$$0 = \tan^2 \theta - 10 \tan \theta + 16$$

$$0 = (\tan \theta - 8)(\tan \theta - 2)$$

$\therefore \tan \theta = 8$ or $\tan \theta = 2$

since $\alpha < \beta$: $\tan \beta = 8$ and $\tan \alpha = 2$. $\checkmark \quad (2)$

some progress

iii) Consider the two paths and find the time travelled to reach P:

Tennis ball 1: $v = 7\sqrt{10}$, $\theta = \beta$, $\tan\beta = 8$ and $x = 10$:

$$t_1 = \frac{10}{7\sqrt{10} \cos\beta}$$

$$= \frac{10 \sec\beta}{7\sqrt{10}} \quad \left(\begin{array}{l} \sec^2\beta = 1 + \tan^2\beta \\ \sec^2\beta = 1 + 8^2 \end{array} \right)$$

$$= \frac{10 \times \sqrt{65}}{7\sqrt{10}}$$

$$t_1 = \frac{\sqrt{650}}{7}$$

✓ ① progress towards a solution.

Tennis ball 2: $v = 7\sqrt{10}$, $\theta = \alpha$, $\tan\alpha = 2$ and $x = 10$:

$$t_2 = \frac{10}{7\sqrt{10} \cos\alpha}$$

$$= \frac{10 \sec\alpha}{7\sqrt{10}}$$

$$\left(\begin{array}{l} \sec^2\alpha = 1 + \tan^2\alpha \\ \sec^2\alpha = 1 + 4 \end{array} \right)$$

$$= \frac{10\sqrt{5}}{7\sqrt{10}}$$

$$= \frac{\sqrt{50}}{7}$$

$$\therefore \text{time elapsed: } t_1 - t_2 = \frac{\sqrt{650} - \sqrt{50}}{7} \text{ seconds.}$$

✓ ②.